SRI JCBM COLLEGE, SRINGERI

DEPARTMENT OF COMPUTER APPLICATIONS

LIBRARY MANAGEMENT SYSTEM

A Project Report

Submitted in partial fulfilment of the requirements for the degree of

BACHELOR OF COMPUTER APPLICATIONS (BCA)

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CERTIFICATE

This is to certify that the project report entitled **“Library Management System”** is a record of project work carried out by the student named on the title page in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA), during the academic year 2026–2027.

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Head of the Department: Dashami Mam

Place: Sringeri Date: 30/10/2026

DECLARATION

I hereby declare that the project work entitled **“Library Management System”** has been prepared by me for academic purposes. The project is a menu-driven console application developed using the C programming language. The work presented in this report is based on the program and its sample execution.

Place: Sringeri
Date: _____

Signature: _____

ACKNOWLEDGEMENT

I express my sincere gratitude to the project guide for guidance, encouragement and valuable suggestions during this project. I thank the Head of the Department and all faculty members of Sri JCBM College, Sringeri, for their support and for providing the opportunity to undertake this work. I also thank my family and friends for their encouragement.

ABSTRACT

The Library Management System is a menu-driven, console-based application developed in the C programming language. It provides an administrator login and supports adding, displaying, searching, issuing, returning and deleting book records. It also calculates a simple overdue fine at a fixed rate of Rs. 2.00 per day. Book information is stored in arrays during program execution. The project demonstrates basic C concepts such as arrays, strings, functions, loops, conditional statements and formatted input/output. Records are held in memory and are not saved permanently after the program exits.

Keywords: C programming, library management, book records, arrays, functions, issue and return, delete book, fine calculation.

TABLE OF CONTENTS

Chapter	Topic	Page
1	Introduction	5
2	Objectives of the Project	5
3	System Study and Analysis	6
4	Feasibility Study	6
5	Hardware and Software Requirements	7
6	System Design	7
7	Implementation	8
8	Testing	9
9	Results and Output	10
10	Advantages and Limitations	11
11	Future Enhancements	11
12	Conclusion	12
13	Bibliography and References	12
Appendix A	Complete C Source Code	13
Appendix B	Sample Outputs	17

Project details to verify before submission: student name, register number, project guide, signatures and academic details.

1. INTRODUCTION

A library maintains information about books and their availability. In a manual system, staff may use registers to record book details and track whether a book has been issued or returned. Searching for a particular book and updating its status can take time.

The Library Management System developed in this project provides a simple computer-based method for maintaining book records. An administrator first logs in using a username and password. After successful login, the administrator can add, display, search, issue, return and delete books. The program also calculates an overdue fine based on the number of days.

This is an introductory console application intended for learning C programming. It does not use a database or permanently store records.

2. OBJECTIVES OF THE PROJECT

The main objectives of the Library Management System are:

• To provide an administrator login before allowing access to the menu.

• To store book ID, title, author and issue status.

• To add new book records and prevent duplicate book IDs.

• To display all books stored in the system.

• To search for a book using its ID.

• To issue an available book and mark it as issued.

• To return an issued book and mark it as available.

• To delete a book record from the catalogue.

• To calculate an overdue fine at Rs. 2.00 per day.

• To practise arrays, strings, functions, loops and decision-making statements in C.

3. SYSTEM STUDY AND ANALYSIS

3.1 Existing System

In a traditional manual library system, book details and issue information are maintained in registers. Searching, checking issue status and deleting outdated records require manual work. This may be time-consuming and can lead to mistakes.

3.2 Proposed System

The proposed system provides a numbered menu through which an administrator performs basic library operations. Book IDs, titles, authors and issue status are stored in arrays. The system searches by book ID and updates the issue status when a book is issued or returned.

3.3 Users of the System

The application is designed for a single administrator or library staff member. The sample login credentials are username “admin” and password “1234”. These credentials are hard-coded for demonstration and are not suitable for real-world security.

3.4 Scope of the System

The project covers basic book catalogue operations and a simple fine calculation. It does not maintain member accounts, borrowing dates, due dates or transaction history.

4. FEASIBILITY STUDY

4.1 Technical Feasibility

The program is written in standard C and can be compiled using GCC or another compatible C compiler. It uses common header files such as stdio.h and string.h. No database or internet connection is required.

4.2 Economic Feasibility

The system can be developed and executed using free C compilers and text editors. An existing desktop or laptop computer is sufficient, so the direct software cost is low.

4.3 Operational Feasibility

The menu-driven interface uses simple numbered options and text prompts. A user with basic computer knowledge can operate the program.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Category	Requirement
Hardware	Desktop or laptop computer, keyboard and display
Operating System	Windows, Linux or macOS
Programming Language	C (C99 or later recommended)
Compiler	GCC or another compatible C compiler
Editor / IDE	VS Code, Code::Blocks, Dev-C++ or any text editor
Additional Software	No database or network software required

The program requires only a basic C development environment. It is suitable for demonstration in a computer laboratory.

6. SYSTEM DESIGN

6.1 Main Modules

Module	Function	Purpose
Login	main()	Checks administrator username and password
Add Book	addBook()	Adds a book with a unique ID
Display Books	displayBooks()	Displays all stored book records
Search Book	searchBook()	Finds a book using its ID
Issue Book	issueBook()	Changes status from available to issued
Return Book	returnBook()	Changes status from issued to available
Delete Book	deleteBook()	Removes a book record from the arrays
Fine Calculation	calculateFine()	Calculates fine at Rs. 2 per overdue day

6.2 Data Design

The program uses parallel arrays. The integer array id[] stores book IDs, the character arrays title[][] and author[][] store book names and authors, and the integer array issued[] stores issue status. A value of 0 means available and a value of 1 means issued. The variable count stores the number of current book records. The maximum capacity is 50 books.

6.3 Menu Design

Choice	Operation
1	Add Book
2	Display Books
3	Search Book
4	Issue Book
5	Return Book
6	Delete Book
7	Calculate Fine
8	Exit

6.4 Fine Formula

Fine = Number of overdue days × Fine rate per day. The fixed fine rate used in this project is Rs. 2.00 per day. For example, 7 overdue days result in a fine of Rs. 14.00.

7. IMPLEMENTATION

7.1 Login Module

The program asks the administrator to enter a username and password. The strcmp() function compares the entered values with the hard-coded credentials “admin” and “1234”. If either value is incorrect, the program displays an error message and terminates.

7.2 Add Book Module

The addBook() function checks whether the catalogue has reached its maximum capacity. It then accepts a book ID and checks for duplicates using findBook(). If the ID is unique, the title and author are stored and the issue status is initialized to 0 (available).

7.3 Display and Search Modules

The displayBooks() function uses a for loop to display each stored book and its status. The searchBook() function accepts a book ID and uses findBook() to locate the matching record.

7.4 Issue and Return Modules

The issueBook() function checks whether the requested book exists and whether it is already issued. If it is available, the status is changed to 1. The returnBook() function checks whether the book exists and is currently issued; if so, it changes the status back to 0.

7.5 Delete Book Module

The deleteBook() function searches for the given book ID. When the record is found, the program shifts all later records one position to the left in each parallel array. The count variable is then reduced by one. If the ID does not exist, the program displays “Book not found!”.

7.6 Fine Calculation Module

The calculateFine() function asks for the number of overdue days. Negative or non-numeric input is rejected. The fine is calculated by multiplying the number of days by the fixed rate of Rs. 2.00 per day.

7.7 C Concepts Used

- Arrays: store multiple book IDs, titles, authors and statuses.
- Strings: store text and compare login details using strcmp().
- Functions: divide the program into separate operations.
- Loops: traverse book records and repeat the menu.
- Conditional statements: validate login and select operations.
- Switch statement: execute the selected menu option.
- Formatted input/output: scanf() and printf() interact with the user.

8. TESTING

The following test cases describe the expected behaviour of the program. They should be run on the compiled program during project demonstration.

Test Case	Input / Action	Expected Result
Correct login	admin / 1234	Login successful and menu displayed
Incorrect login	Invalid username or password	Error message and program exits
Add book	Enter a new ID, title and author	Book added successfully
Duplicate ID	Enter an existing book ID	Duplicate ID message
Display books	Choose option 2	All stored book details are displayed
Search existing	Search for a stored ID	Book details are displayed
Search missing	Search for an unknown ID	Book not found message
Issue book	Issue an available book ID	Status changes to Issued
Issue twice	Issue the same book again	Book already issued message
Return book	Return an issued book ID	Status changes to Available
Delete book	Delete an existing book ID	Book deleted successfully
Fine calculation	Enter 7 overdue days	Fine displayed as Rs. 14.00
Exit	Choose option 8	Thank you message and program terminates

9. RESULTS AND OUTPUT

The program provides a working menu for basic library record management. After successful login, the administrator can add, display, search, issue, return and delete book records. It also calculates a fine from the number of overdue days.

9.1 Sample Fine Calculation

Input	Calculation	Result
7 overdue days	7 × Rs. 2.00	Rs. 14.00

9.2 Result Summary

The sample execution in Appendix B demonstrates successful login, adding a book, displaying and searching the record, issuing and returning the book, deleting a record, and calculating a fine. Actual output depends on the values entered by the user.

10. ADVANTAGES AND LIMITATIONS

10.1 Advantages

• Simple and easy-to-use menu-driven interface.
• Supports adding, displaying and searching book records.
• Prevents duplicate book IDs.
• Tracks whether each book is issued or available.
• Supports returning and deleting book records.
• Calculates a simple overdue fine automatically.
• Demonstrates important C programming concepts.

10.2 Limitations

• Book records are stored only in memory and are lost when the program ends.
• Titles and author names are limited to one word in this version.
• The program does not maintain member details or borrowing history.
• Fine calculation is not connected to actual issue and due dates.
• Only one hard-coded administrator login is supported.
• The password is stored in the source code and is not secure for real deployment.

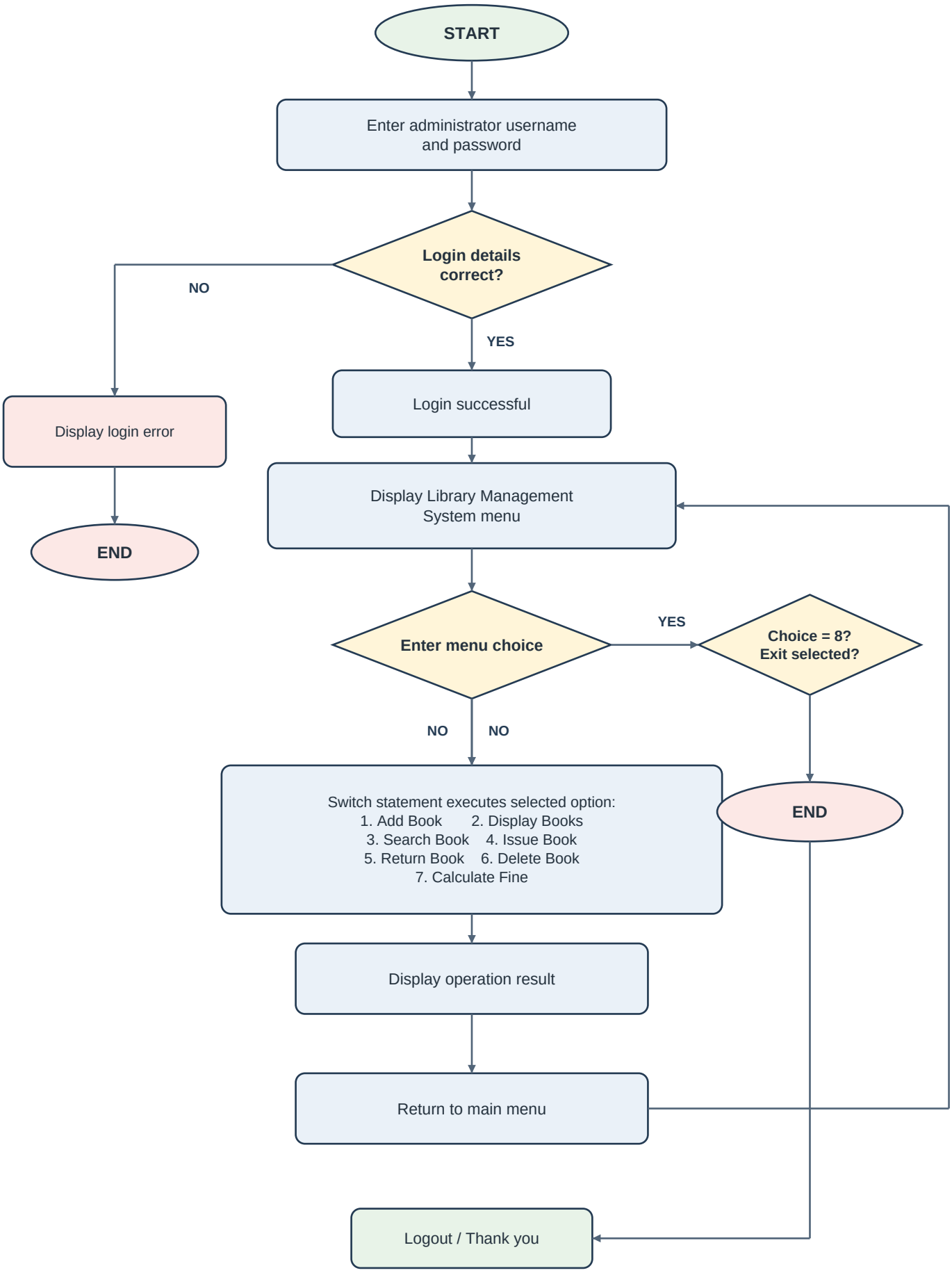
11. FUTURE ENHANCEMENTS

The project can be improved in several ways:

- Use file handling or a database to save book records permanently.
- Allow book titles and author names containing spaces.
- Add member registration and member login facilities.
- Store issue dates, due dates and return dates.
- Calculate fines automatically from overdue transactions.
- Add safer password handling and multiple user roles.
- Generate reports for issued, returned and available books.
- Add confirmation before deleting a book record.
- Develop a graphical user interface for easier operation.

FLOWCHART OF LIBRARY MANAGEMENT SYSTEM

Program flow based on the modules and menu options described in this report



Note: Book operations and fine calculation return to the menu; option 8 terminates the program.

12. CONCLUSION

The Library Management System is a basic C programming project that demonstrates how a menu-driven application can be used to manage book records. It provides administrator login and supports adding, displaying, searching, issuing, returning and deleting books. It also calculates an overdue fine using a fixed daily rate. The project gives practical experience with arrays, strings, functions, loops, conditional statements and formatted input/output. Permanent storage, member records and stronger security can be introduced in future versions.

13. BIBLIOGRAPHY AND REFERENCES

1. Kernighan, B. W., and Ritchie, D. M. (1988). *The C Programming Language*, 2nd Edition. Prentice Hall.

2. Balagurusamy, E. *Programming in ANSI C*. McGraw-Hill Education.

3. GNU Project. GCC Online Documentation: https://gcc.gnu.org/onlinedocs/

4. cppreference.com. C language and standard library reference: https://en.cppreference.com/w/c

APPENDIX A: COMPLETE C SOURCE CODE

The following complete program includes login, add, display, search, issue, return, delete and fine calculation operations. Default login: username **admin**; password **1234**.

```
#include <stdio.h>
#include <string.h>

#define MAX_BOOKS 50
#define FINE_PER_DAY 2.0

int id[MAX_BOOKS], issued[MAX_BOOKS];
char title[MAX_BOOKS][50], author[MAX_BOOKS][50];
int count = 0;

int findBook(int bookId);
void addBook(void);
void displayBooks(void);
void searchBook(void);
void issueBook(void);
void returnBook(void);
void deleteBook(void);
void calculateFine(void);

int main(void)
{
    char username[20], password[20];
    int choice;

    printf("===== LIBRARY MANAGEMENT SYSTEM =====\n");
    printf("\n===== LIBRARY LOGIN =====\n");

    printf("Username: ");
    scanf("%19s", username);

    printf("Password: ");
    scanf("%19s", password);

    if (strcmp(username, "admin") != 0 ||
        strcmp(password, "1234") != 0)
    {
        printf("Wrong username or password!\n");
        return 0;
    }

    printf("Login successful!\n");

    do
    {
        printf("\n===== LIBRARY MENU =====\n");
        printf("1. Add Book\n");
        printf("2. Display Books\n");
        printf("3. Search Book\n");
        printf("4. Issue Book\n");
        printf("5. Return Book\n");
        printf("6. Delete Book\n");
        printf("7. Calculate Fine\n");
        printf("8. Exit\n");
        printf("Enter choice: ");

        if (scanf("%d", &choice) != 1)
        {
            printf("Invalid input!\n");
            int ch;
            while ((ch = getchar()) != '\n' && ch != EOF) { }
            choice = 0;
            continue;
        }

        switch (choice)
```

```
        {
            case 1:
                addBook();
                break;
            case 2:
                displayBooks();
                break;
            case 3:
                searchBook();
                break;
            case 4:
                issueBook();
                break;
            case 5:
                returnBook();
                break;
            case 6:
                deleteBook();
                break;
            case 7:
                calculateFine();
                break;
            case 8:
                printf("Thank you!\n");
                break;
            default:
                printf("Invalid choice!\n");
        }

    } while (choice != 8);

    return 0;
}

int findBook(int bookId)
{
    int i;
    for (i = 0; i < count; i++)
    {
        if (id[i] == bookId)
            return i;
    }
    return -1;
}

void addBook(void)
{
    int bookId;

    if (count >= MAX_BOOKS)
    {
        printf("Library is full!\n");
        return;
    }

    printf("Enter book ID: ");
    if (scanf("%d", &bookId) != 1)
    {
        printf("Invalid book ID!\n");
        int ch;
        while ((ch = getchar()) != '\n' && ch != EOF) { }
        return;
    }

    if (findBook(bookId) != -1)
```

```
    {
        printf("Book ID already exists!\n");
        return;
    }

    id[count] = bookId;

    printf("Enter title (one word): ");
    scanf("%49s", title[count]);

    printf("Enter author (one word): ");
    scanf("%49s", author[count]);

    issued[count] = 0;
    count++;

    printf("Book added successfully!\n");
}

void displayBooks(void)
{
    int i;

    if (count == 0)
    {
        printf("No books available!\n");
        return;
    }

    printf("\n===== BOOK DETAILS =====\n");
    for (i = 0; i < count; i++)
    {
        printf("\nBook ID: %d\n", id[i]);
        printf("Title: %s\n", title[i]);
        printf("Author: %s\n", author[i]);

        if (issued[i] == 1)
            printf("Status: Issued\n");
        else
            printf("Status: Available\n");
    }
}

void searchBook(void)
{
    int bookId, index;

    printf("Enter book ID to search: ");
    if (scanf("%d", &bookId) != 1)
    {
        printf("Invalid book ID!\n");
        int ch;
        while ((ch = getchar()) != '\n' && ch != EOF) { }
        return;
    }

    index = findBook(bookId);

    if (index == -1)
    {
        printf("Book not found!\n");
    }
    else
    {
        printf("Book found!\n");
```

```
        printf("Title: %s\n", title[index]);
        printf("Author: %s\n", author[index]);
        printf("Status: %s\n",
               issued[index] == 1 ? "Issued" : "Available");
    }
}

void issueBook(void)
{
    int bookId, index;

    printf("Enter book ID to issue: ");
    if (scanf("%d", &bookId) != 1)
    {
        printf("Invalid book ID!\n");
        int ch;
        while ((ch = getchar()) != '\n' && ch != EOF) { }
        return;
    }

    index = findBook(bookId);

    if (index == -1)
        printf("Book not found!\n");
    else if (issued[index] == 1)
        printf("Book already issued!\n");
    else
    {
        issued[index] = 1;
        printf("Book issued successfully!\n");
    }
}

void returnBook(void)
{
    int bookId, index;

    printf("Enter book ID to return: ");
    if (scanf("%d", &bookId) != 1)
    {
        printf("Invalid book ID!\n");
        int ch;
        while ((ch = getchar()) != '\n' && ch != EOF) { }
        return;
    }

    index = findBook(bookId);

    if (index == -1)
        printf("Book not found!\n");
    else if (issued[index] == 0)
        printf("Book was not issued!\n");
    else
    {
        issued[index] = 0;
        printf("Book returned successfully!\n");
    }
}

void deleteBook(void)
{
    int bookId, index, i;

    printf("Enter book ID to delete: ");
    if (scanf("%d", &bookId) != 1)
```

```
    {
        printf("Invalid book ID!\n");
        int ch;
        while ((ch = getchar()) != '\n' && ch != EOF) { }
        return;
    }

    index = findBook(bookId);

    if (index == -1)
    {
        printf("Book not found!\n");
        return;
    }

    for (i = index; i < count - 1; i++)
    {
        id[i] = id[i + 1];
        strcpy(title[i], title[i + 1]);
        strcpy(author[i], author[i + 1]);
        issued[i] = issued[i + 1];
    }

    count--;
    printf("Book deleted successfully!\n");
}

void calculateFine(void)
{
    int days;
    float fine;

    printf("Enter number of overdue days: ");
    if (scanf("%d", &days) != 1 || days < 0)
    {
        printf("Invalid number of days!\n");
        int ch;
        while ((ch = getchar()) != '\n' && ch != EOF) { }
        return;
    }

    fine = days * FINE_PER_DAY;

    printf("Fine rate: Rs. %.2f per day\n", FINE_PER_DAY);
    printf("Total library fine: Rs. %.2f\n", fine);
}
```

APPENDIX B: SAMPLE OUTPUTS

The following console sessions are illustrative. Each operation is shown separately for clarity; the program returns to the menu after each operation.

B.1 Successful Login and Menu

```
===== LIBRARY MANAGEMENT SYSTEM =====

===== LIBRARY LOGIN =====
Username: admin
Password: 1234
Login successful!

===== LIBRARY MENU =====
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Calculate Fine
8. Exit
Enter choice:
```

B.2 Adding and Displaying a Book

```
Enter choice: 1
Enter book ID: 101
Enter title (one word): WingsOfFire
Enter author (one word): AbdulKalam
Book added successfully!

Enter choice: 2

===== BOOK DETAILS =====
Book ID: 101
Title: WingsOfFire
Author: AbdulKalam
Status: Available
```

B.3 Searching for a Book

```
Enter choice: 3
Enter book ID to search: 101
Book found!
Title: WingsOfFire
Author: AbdulKalam
Status: Available
```

B.4 Issuing and Returning a Book

```
Enter choice: 4
Enter book ID to issue: 101
Book issued successfully!

Enter choice: 2

===== BOOK DETAILS =====
Book ID: 101
Title: WingsOfFire
Author: AbdulKalam
Status: Issued

Enter choice: 5
Enter book ID to return: 101
Book returned successfully!
```

B.5 Deleting a Book

```
Enter choice: 6
Enter book ID to delete: 101
Book deleted successfully!

Enter choice: 2
```

```
No books available!
```

B.6 Fine Calculation

```
Enter choice: 7
Enter number of overdue days: 7
Fine rate: Rs. 2.00 per day
Total library fine: Rs. 14.00
```

B.7 Exiting the Program

```
Enter choice: 8
Thank you!
```

B.8 Incorrect Login

```
===== LIBRARY LOGIN =====
Username: student
Password: wrong
Wrong username or password!
```

PROJECT DETAILS TO COMPLETE BEFORE SUBMISSION

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